GCF-NSD Lab Request Package v0.2

External feasibility request package - safe disclosure version

Purpose and Scope

This package is intended for technical feasibility discussion with a steel laboratory or materials R&D partner. It asks whether a weldable HSLA-like marine/offshore structural steel matrix can support coupon-level routes that reduce coupled failure risk: localized corrosion/pitting to fatigue crack initiation, crack-path connectivity, and weld/HAZ amplification.

This is not a final alloy composition, commercial specification, certified performance claim, or military-grade claim. Improvements over the experimental baseline should be interpreted only as feasibility signals unless later validated by expanded testing.

Baseline Control

The baseline must not be selected only for convenience or to maximize apparent improvement. A dual-baseline logic is recommended.

Baseline	Role	Recommended examples	Interpretation rule
B0-A	Physical first-round coupon baseline	API 2W Grade 50 or API 2H Grade 50; if unavailable, ASTM A131 / ABS DH36 or EH36	Improvement vs. B0-A = feasibility signal only
B0-B	Literature/strategic reference baseline	HSLA-80 / HSLA-100-class naval structural steel references	Do not imply superiority unless directly tested under comparable conditions

Candidate Sample Groups

Group	Role	Description
B0-A	Commercial baseline	Available marine/offshore HSLA-like structural steel.
S1	CA-biased candidate	Route intended to improve crack arrest, crack deflection, crack blunting, crack branching, or crack-path tortuosity.
S2	CF-biased candidate	Route intended to reduce pitting-to-fatigue transition, localized corrosion sensitivity, or corrosion-fatigue coupling.
S3	CA-CF candidate	Route intended to combine CA behavior and CF behavior without sacrificing weldability or HAZ stability.
S4	Optional HAZ/process-adjusted CA-CF	Triggered only if S3 shows or is expected to show HAZ/weld instability.

Candidate Technical Direction Examples - Non-Recipe Level

The following examples are concept families only. They are provided to help the laboratory respond with feasible routes. They are not alloy recipes, element ranges, heat-treatment parameters, or process instructions.

Functional direction	Potential concept families	Disclosure boundary
CA-oriented	Acicular ferrite or bainitic packet refinement; fine-grained or multi-phase crack-deflection structures; interface/inclusion-shape control; microstructural barriers to straight crack propagation.	Do not provide exact composition, heat-treatment time/temperature, precipitation target recipe, or proprietary processing route before NDA.
CF-oriented	Localized corrosion/pitting resistance improvement; microstructure uniformity	Discuss mechanism families and feasibility only. Avoid detailed surface-treatment

	around susceptible interfaces; surface or near-surface treatment concepts; reduced pit-to-crack transition sensitivity.	sequence or chemistry.
CA-CF combined	Balanced crack-path tortuosity plus pit-initiation suppression; HAZ-compatible processing route; balanced microstructure rather than maximum hardness/strength.	Candidate route must preserve weldability and HAZ behavior. Do not treat higher hardness as success by itself.

S4 Trigger Rule

S4 should not be added casually. It is triggered only under one of the following conditions:

- The laboratory expects S3 to create weld/HAZ instability based on known metallurgical constraints.
- Tier 1 hardness mapping or microstructure suggests HAZ sensitivity.
- S3 shows positive CA/CF behavior in base metal but negative HAZ-like behavior.
- The laboratory believes a process-adjusted route could preserve CA-CF effects while reducing HAZ risk.

Coupon Conditions

Each material group should include at minimum a base-metal coupon and a simulated HAZ coupon or welded-region proxy coupon. This prevents a false positive in which the base metal looks promising but the weld/HAZ condition collapses.

Group	Base metal coupon	Simulated HAZ or weld-region proxy
B0-A	Required	Required
S1 CA	Required	Required
S2 CF	Required	Required
S3 CA-CF	Required	Required
S4 optional	Optional	Required if included

Suggested Test Condition Variables

The laboratory should propose realistic test conditions based on equipment, standards, and expected service environment. The following variables should be specified in any quotation or test plan:

- Corrosive environment: artificial seawater, 3.5 wt% NaCl, or other service-relevant medium.
- Temperature: ambient or service-relevant low-temperature condition where appropriate.
- Corrosion pre-exposure duration and surface condition before fatigue testing.
- Cyclic loading stress ratio R, frequency, waveform, and stress/amplitude level.
- Dissolved oxygen, pH, and aeration control if feasible.
- Definition of failure/end point: crack initiation, fatigue life, crack growth rate, or morphological threshold.
- Repeat count per condition and statistical limitations of any first-round result.

No single condition is mandated in this request package. The purpose is to obtain a laboratory-proposed, defensible condition set rather than impose a premature specification.

Tiered Test Menu

Tier 1 - Low-cost screening

- Microstructure observation
- Hardness mapping, especially across HAZ-like regions
- Basic tensile properties
- Basic corrosion exposure and pit density/depth observation
- Impact or toughness proxy if feasible

Tier 2 - Core feasibility tests

- Fatigue after corrosion pre-exposure

- Crack initiation cycle comparison
- Percentage of cracks initiating from pits
- Crack path tortuosity, deflection, branching, and connectivity
- Fatigue crack growth proxy or direct da/dN if feasible
- Fracture surface observation
- Same comparisons under simulated HAZ or welded-region proxy conditions

Tier 2 optional - Low-temperature toughness / fracture toughness

- Charpy impact at baseline-relevant low temperature if feasible
- CTOD or fracture toughness proxy for promising candidates or HAZ-like coupons
- These are not required for every first-round coupon if budget is limited, but should be considered before any upgrade claim

Priority Order Under Budget Constraints

1. B0-A vs. S3 CA-CF under corrosion pre-exposure plus fatigue or crack-initiation proxy.

2. B0-A vs. S3 CA-CF under simulated HAZ or welded-region proxy condition.

3. S1 vs. S2 vs. S3 crack-path morphology comparison.

4. Low-temperature Charpy or CTOD consideration for promising candidates and HAZ-like coupons.

5. Expanded repeat coupons or more rigorous fatigue crack growth testing.

Interpretation Rules

Signal type	Definition	Action
Positive	S3 improves or does not degrade fatigue after corrosion, reduces/delays pit-origin crack initiation, improves crack-path tortuosity/deflection/connectivity, and does not degrade under HAZ-like condition.	Keep S3 active as primary feasibility candidate.
Boundary	Static corrosion improves but fatigue after corrosion does not; base metal improves but HAZ worsens; S3 does not outperform S1/S2; only isolated coupons improve.	Do not upgrade; refine test or candidate route.
Negative	S3 is worse than B0-A in corrosion-fatigue, HAZ-like coupon degrades, crack paths become straighter/faster/more connected, or improvements are only hardness/strength effects.	Stop or downgrade S3.

Requested Lab Feedback

1. Which commercial steel grade would you recommend as B0-A, and why?

2. Can API 2W Grade 50, API 2H Grade 50, ABS DH36, or ABS EH36 be sourced or approximated?

3. What concept-family routes could produce CA-biased behavior without becoming a recipe disclosure?

4. What concept-family routes could produce CF-biased behavior without becoming a recipe disclosure?

5. Is a CA-CF combined route technically plausible without harming weldability or HAZ stability?

6. Which concept families would you reject immediately as too brittle, too costly, too weld-sensitive, or too hard to scale?

7. Can simulated HAZ coupons or welded-region proxy coupons be prepared?

8. What environmental conditions would you propose for corrosion pre-exposure and corrosion-fatigue screening?

9. What stress ratio, frequency, waveform, and loading level would you recommend for first-round fatigue screening?

10. Can low-temperature Charpy, CTOD, or a fracture-toughness proxy be included economically?

11. What is the minimum practical Tier 1 package, cost, and timeline?

12. What is the minimum practical Tier 1 + limited Tier 2 package, cost, and timeline?

13. What result would be sufficient to justify a second-round expanded study?

14. What risks would make this concept impractical?

15. What information is needed before formal quotation?

16. What items should remain under NDA before route-level discussion?

Confidentiality and Disclosure Boundary

This document is intended only for technical feasibility discussion. It does not disclose final alloy composition, heat-treatment recipe, proprietary scoring method, internal inference framework, commercial deployment plan, or certified naval/military performance claim. Before exchanging detailed candidate routes, experimental data, or patent-relevant implementation details, a confidentiality agreement is recommended.

Version Note

v0.2 adds non-recipe candidate concept families, suggested test condition variables, low-temperature Charpy/CTOD consideration, and an explicit S4 trigger rule. It remains a safe external request package and should not be treated as a detailed materials recipe.